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Smart farming revolution: Leveraging machine learning for sustainable agriculture

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HIGHLIGHTS

- CNN, RF, and LSTM analyze spatial, temporal, and visual farm data.
- Data from 847 farms supports precision farming and climate resilience.
- ML boosts yield by 20 %, cuts water by 15 %, and saves $5K in pesticides.
- CNN detects crop disease; LSTM forecasts trends and risks.
- Scalable ML bridges traditional and smart farming practices.

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ABSTRACT

Problem: In today's world, agriculture faces several challenges, including environmental degradation, crop diseases, and ineffective resource utilization. These challenges affect sustainability as well as food security.
Method: ology: In order to address the above-mentioned problems, several machine learning (ML) approaches, namely, Convolutional Neural Network (CNN), Random Forest (RF), Long Short Term Memory (LSTM), and other deep neural models, were used. These techniques aid in maximizing resource allocation, improving climate resilience, and enhancing early disease detection in agricultural ecosystems. Data for this study were collected through a questionnaire survey from 847 farmers across three main regions of China and were filtered using random sampling. Moreover, the collected data were analyzed through the Statistical Package for the Social Sciences (SPSS).
Results: The ML-based technique improved productivity, reduced operational costs, and enhanced sustainability. It attained a 20 % crop yield increase, minimized water consumption by 15 %, and reduced pesticide costs by $5000 per year in comparison to conventional farming practices.
Impact: This study shows the ability of ML technologies to revolutionize traditional agriculture, fostering stronger and sustainable agriculture systems that can sustain contemporary demands without compromising ecological and economic stability.

1. Introduction

Agriculture plays a fundamental role in sustaining human life by ensuring food production and contributing to economic development (Guo et al., 2024). In recent years, Climate change, resource scarcity, as well as growing demand for food are the crucial challenges faced by agriculture (Dash et al., 2022). Although they were successful in the past, conventional agricultural techniques often lead to unsustainable environmental effects, excessive chemical input, and inefficient resource use (de Paula Albuquerque et al., 2024). The demand for modern agriculture, such as food safety guaranteeing, risk prediction associated with climate change, and resource management optimization, is difficult for such approaches to adapt (Wanyama et al., 2024; Lakhouit et al., 2025). Furthermore, traditional methods find it difficult to allocate resources and manage crops in an era of rapid increase in population and variations in the climate, both of which are essential for sustaining the productivity of agriculture (Vinod Chandra et al., 2024).

As noted by Krasilnikov et al. (2022), traditional farming practices

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face challenges such as higher water consumption, higher chemical usage, and more time-intensive, etc. To address the challenges faced by traditional farming, innovative technologies like big data and Machine Learning (ML) were created. The world's foremost country in cropland intensification, China, has significantly adopted and advanced these technologies (Zhai et al., 2024). The main important technique in this phase is precision farming (Costa et al., 2023). Here, drones, Artificial Intelligence (AI), and Internet of Things (IoT) are used to monitor soil health, weather, and moisture in real time. This assists in enhancing crop yield and the decision-making process (Arya, 2024). Yousaf et al. (2023) state that smart farming results in high-quality crops with low-cost, labor-minimal farming. Reduction in resource waste, carbon footprint, and chemical input is also aided largely by ML technologies. With the merging of precision agriculture and ML techniques, economic viability is achieved along with ecological responsibility (Arora et al., 2022).

This research focused on China's farmland due to its complex, multi-dimensional, and symbolic nature of global agricultural challenges. China has several kinds of crops, climatic zones, and constraining resources, and therefore, it is a suitable region to develop and validate scalable and flexible ML models. China's problems in agriculture were water scarcity, land degradation, and climate change. These are not problems that are specifically faced in China but are being faced across the world. This means that the lessons learned from addressing water scarcity and climate variability in China strengthen the global initiative, which aids in sustainable farming practices. Several studies have been conducted on this topic, but they employ limited numbers of ML methods, oftentimes neglecting critical aspects like scalability, flexibility, and eco-friendliness. This research fills this important research gap by suggesting a low-cost and scalable ML framework for sustainable agriculture. The principal research problem is the lack of comprehensive ML models that, at the same time, optimize productivity, environmental sustainability, and resource use efficiency across different farming regimes. The suggested research suggests a resilient ML-enabled system that enhances agricultural efficiency by data-driven decision support for resource management, plant disease detection, and climate resilience. It employs a broad range of algorithms, including Random Forest (RF), Convolutional Neural Networks (CNNs), Long Short-Term Memory Networks (LSTMs), eXtreme Gradient Boosting (XGBoost), Support Vector Machines (SVMs), Decision Trees, K-Nearest Neighbors (KNN), and Transformer models to successfully resolve a range of agricultural issues. RF, CNNs, and LSTMs are used particularly because they perform better on structured data, image-based analysis, and time-series prediction, respectively. Model choice is determined by the balance of task performance, computation cost, and scalability across various agri-setting scenarios. Although more advanced models such as XGBoost and Transformers are used in situations with high-accuracy demands or sequence modeling complexity, their computational cost might render them unsuitable for real-time or limited agricultural applications. Their application is hence selectively intended to fit specific contexts in a bid to possess pragmatism. As opposed to the traditional practices, which lacked good scalability and were expensive to adopt, the framework proposed herein complies with the contemporary farming practices and is affordable, in addition to being flexible. With the synchronization of model selection with task requirement and availability of resources, and by resolving conventional agricultural practices using modern ML approaches, this solution provides a conciliatory, responsive, and resilient intelligent agriculture solution with long-term productivity and environmental conservation.

1.1. Contribution and novelty of the study

The key contribution of the study is.

- ***Optimization of Resources:*** Leverages ML algorithms that track past and real-time data to optimize water, fertilizer, and pesticide use, reducing waste and greenhouse gas emissions, and provide predictive irrigation scheduling.
- ***Early Disease Detection:*** Enables the pre-diagnosis of crop disease through image identification methods with an early intervention to restrict the losses, and integrates ML systems into the conventional farm practice to promote technology adoption.
- ***Economic Sustainability:*** Demonstrates significant cost savings through optimized resource allocation and improved supply chain efficiency, giving farmers predictive insights to make better financial decisions on crop development and market conditions
- ***Data Privacy and Job Creation:*** Introduces a robust framework for the protection of farmer data by privacy-driven measures and the creation of new employment opportunities in information technology and data science sectors, alleviating rural-urban gaps between traditional farming and innovation-age activities.
- ***Climate Resilience and Sustainable Practices:*** Synthesizes ML with climate policy to enable climate variability resilience, forecast soil erosion hazard for better soil fertility, and develop ML-based irrigation management systems that initiate sustainable water utilization and farm productivity.

All of these points together envision the revolutionary potential of proposed research in achieving sustainable agriculture practices through the convergence of ML.

The rest of this paper is organized as follows: Section 2 overviews related works and defines research gaps, Section 3 overviews the theoretical framework, Section 4 overviews the conceptual framework along with the six hypotheses, Section 5 gives an overview of the methodology, Section 6 provide experimental results, Section 7 overviews an extensive discussion along with theoretical and practical implications, and finally Section 8 concludes the paper and proposes future directions of research.

2. Literature review

The current literature review presents the existing literature studies in the field of agriculture based on the use of ML approaches.

2.1. Crop selection and yield prediction

There have been recent developments on crop yield prediction focused on hybrid and explainable machine learning (ML) models for improving their accuracy and robustness to dynamic environmental conditions. Kemmannu et al. (2024) developed an Adaptive Neuro-Fuzzy Inference Systems (ANFIS) model in conjunction with XGBoost. The intention was to create a model that utilized real-time climate and soil conditions to make robust predictive forecasts in an environment of agricultural uncertainty. Sahoo et al. (2024) worked on ML models that used climate uncertainties to examine and analyze rice yield gaps, emphasizing localized, data-driven decisions, based on agricultural emissions that improve food security. Kheir et al. (2025) developed a modeling framework by merging remote sensing with process-based models and ML techniques to produce wheat yield predictions in a geographically explicit manner.

2.2. Resource allocation

Agricultural resource planning refers to the efficient and well-structured allocation of fundamental inputs like water, fertilizers, pesticides, and manpower, intending to impart productivity and sustainability to the maximum possible degree. ML models are considerably suitable in aiding such decisions through rigorous scrutiny of intricate relationships among different factors of inputs. Singh (2022) discussed optimal strategies for water and land resource utilization in agriculture and suggested policy-level interventions for sustaining agricultural output. The research employed simulation models to consider long-term

environmental effects identified regional differences in resource supply and demand, and built resilience in the future through the integration of resource management. Taheri et al. (2024) reviewed the optimization methods for cropping pattern optimization, categorizing techniques according to objective functions like profit, yield, and sustainability. They described linear, nonlinear, and metaheuristic decision-making methods and identified problems with multi-objective optimization under uncertainty. The review also found the potential of soft computing in precision agriculture. Karimi et al. (2024) optimized cropping patterns across Ilam province based on sustainability-oriented modeling combining climate, soil, and water information for regional-scale planning. Their model enhanced efficiency in the use of resources while avoiding environmental loss, employed mathematical programming to inform agricultural policy, and showed the capability of decision-support tools in regional farming systems. Ahmad and Zhang (2022) suggested an optimization framework for mixed groundwater–surface systems for agricultural water management based on hydrological models that control the resource availability and demand in agriculture. They suggested the significance of sustainable withdrawals and recharge rates, validated their research in a semi-arid catchment under water-shortage conditions, and offered practical implications for water-saving agriculture practice.

2.3. Supply chain and logistics optimization

Agricultural supply chains are comprised of multifaceted networks with diverse stakeholders and adaptive decision-making processes, from the harvest to storage, transportation, and distribution to markets. Li et al. (2024) have developed a Sustainable Environmental Index (SEI) for efficient management of agricultural land with a combination of logistics, environmental, and social indicators for decision-making assistance for supply chain segmentation. The index was also used for real-world datasets in regional crop logistics planning and assisted policymakers in coordinating productivity with environmental health. Wu et al. (2022) also presented a new routing algorithm to minimize transit time and maintain it in a fresh state while in transit. The model also possessed constraints like vehicle capacity and delivery windows, and was simulated with lower operating costs and therefore suitable for rural market last-mile delivery. Rajabzadeh and Fatorachian (2023) compared the use of IoT technology in agricultural logistics to a conceptual framework, defining barriers and drivers of IoT implementation in farm-to-market value chains. They highlighted digitalization as a force for supply chain efficiency improvement and debated strategic considerations for smart agriculture in the frontier markets, employing survey-based empirical evidence for logistics companies. Gharye Mirzaei et al. (2024) developed an uncertainty-level mathematical model for maximizing fresh date fruit logistics with financial and environmental risks maximized for supply chain decision-making. Their stochastic model method considered perishability and variability in demand and was empirically validated by case studies conducted in the Middle East to encourage risk-conscious planning for agricultural supply chain resilience.

2.4. Irrigation and environmental control

As the climate has been fluctuating, the use of smart environmental control systems has become more crucial to sustainable agriculture. Lee et al. (2025) developed an automated irrigation system that avoids wastage and enhances irrigation efficiency. Real-time field tests validate and test the system. This helps in recording meaningful water savings as well as crop limits and meteorological sensor utilization. It was proven to scale across various crops and sites. Farooq et al. (2022) introduced an IoT and fuzzy logic-based intelligent greenhouse management system wherein irrigation and climatic conditions were managed automatically based on real-time input. This minimized human intervention, enhanced yield performance, and underscored the potential of edge computing for speedy decision-making, augmented with pilot deployment in a controlled agrifacility. Abuzanouneh et al. (2022) introduced an ML-based irrigation prediction model using sensor fusion, collecting soil moisture, temperature, and humidity data to enable targeted watering and reduce resource waste. The model provided real-time decision support via mobile app integration, promoting sustainable water use in arid zones. Sheline et al. (2024) evaluated the controller for resource-limited agriculture, focusing on cost-effective hardware-software integration and analyzing system responsiveness under low-connectivity conditions. It enabled smart irrigation through predictive scheduling, emphasizing its potential for adoption by smallholder farmers. Munaganuri and Rao (2024) utilized multimodal image processing to estimate irrigation demand, analyzing vegetation indices and thermal imagery to detect water stress. Their approach enabled site-specific irrigation recommendations and improved seasonal and daily water use planning, advancing remote sensing for precision irrigation.

2.5. Privacy and federated learning (FL)

Data sharing among agricultural stakeholders offers significant benefits for improving decision-making and operational efficiency, but it also introduces serious privacy concerns. Jararweh et al. (2023) reviewed enablers of smart agriculture, emphasizing edge, cloud, and blockchain technologies, while addressing data privacy, latency, and trust challenges in data sharing. They advocated FL to avoid raw data transmission and discussed secure AI architectures, highlighting interoperability between smart devices and learning platforms. Behera et al. (2025) developed their version of FL to predict crop disease using deep learning (DL), maintaining privacy, training within a multi-party environment, while also providing reasonably good performance despite the data not being uniformly distributed. This promotes more effective, timely disease intervention and alleviates the genuine concerns with collaborative AI. Similarly, Friha et al. (2022) introduced a federated learning-based intrusion detection system (FELIDS) in the agricultural IOT context, for enhanced cybersecurity via distributed anomaly detection across devices. This showed the system performed well in adversarial conditions. Li et al. (2024) developed pruning in FL for crop yield forecasts in agriculture, in communication and computation resource considerations, while preserving privacy, making the approach viable for low-power, remote settings.

2.6. Early detection and integration with traditional practices

An important aspect of sustainable agriculture is the early identification of crop disease, pest infestation, and environmental stress, since almost every yield loss begins here. ML models, particularly CNN and SVM, are set to achieve a great degree of success in crop disease early identification with hyperspectral and drone images. This model identifies stress or infection in a plant before any physical visual signs, which allows intervening sooner and in fewer instances to use pesticides. Vats et al. (2024) introduced Neural Network with RF Framework (NeuroRF) FarmSense using IoT and neural networks as a better way to care for crops by identifying early signs of stress and disease. Joshi et al. (2025) developed a tomato disease diagnosis ML model, employing DL through hybrid data augmentation that generalized extremely well and produced robust performance across diverse public datasets. Patra et al. (2024) created DigiFarmer, an ML-enabled platform for sustainable crop management, that supplied useful insights for crop diagnostics and pest management. Rani et al. (2022) concentrated on automated weed detection with visual identification via computer vision-based techniques to remove weeds selectively while reducing pesticide consumption and monoculture in a smart farming ecosystem. Finally, Mohanty et al. (2025) designed a Robotic sprayer using You Only Look Once version 5 (YOLOv5) that precisely locates and sprays herbicide only on weeds occurring in rice fields, thereby minimizing chemical waste using

real-time detection and classification. Collectively, these innovations illustrate the measurable way by which adventure in AI and ML brings substantial value to modern agriculture, allows precision farming, improved efficiencies, and endorses sustainable practices.

2.7. Interoperability and Job Creation

Interoperability in intelligent agricultural systems enhances data integration, data communication, and decision-making across several platforms, sensors, and stakeholders. It improves not only the efficiency and accuracy of systems, but also is a source of new job opportunities, especially concerning agri-tech infrastructure design, deployment, and management. The present literature also highlights the "interpretability" of models to show trust and transparency. This also encourages broader usage of AI technologies in agriculture. For example, Paudel et al. (2023) characterized the interpretability of the decision-making of the DL framework for predicting crop yields. This model used Shapley additive explanations (SHAP) values to differentiate how input features impacted yield predictions, producing a level of transparency that increases user trust in the end-use of the system. This approach also contributes to the interoperability of AI mo dels and existing agricultural decision support systems, which is a significant benefit to operationalized AI applications in agriculture. Obster et al. (2023) examined interpretable boosting algorithms and applied Explainable Gradient Boosting Machines (EGBMs) to complex environmental and agricultural datasets. Ultimately, this study provided several examples of how the interpretation of ML provides the opportunity for data scientists to connect with agricultural practitioners and provide reasonable,

Table 1
Comparative analysis of ML models for agricultural applications.

Model	Accuracy	Computational Cost	Scalability	Strengths	Drawbacks	Adoption Cost	Infrastructure Needs	Smallholder Suitability
CNN	High (esp. for image-based tasks)	High (requires GPUs and large datasets)	Moderate to High	Excellent for image recognition (e.g., disease detection)	Needs extensive labeled data and computational resources	High	High-performance GPUs, imaging devices	Low to Moderate
SVM	High for small- to medium-sized datasets	Moderate	Low	Performs well with high-dimensional data, interpretable	Less effective on large-scale datasets	Low	Standard computers	Moderate
RF	Moderate to High	Low to Moderate	High	Robust to noise, interpretable, good baseline model	Slower inference with many trees	Low	Basic computing resources	High
XGBoost	High	Moderate to High	High	Handles missing data, high predictive power	Hyperparameter tuning is required, sensitive to noise	Moderate	Multi-core processors recommended	Moderate
ANFIS (with XGBoost)	High	High	Moderate	Adaptive, explainable hybrid model suited to environmental variability	Complex integration, high training cost	High	Requires expert calibration and a hybrid infrastructure	Moderate
SHAP + Deep Learning	High with interpretability	High	Moderate	Improves model transparency and user trust	Computationally intensive explanations	Moderate	DL frameworks + interpretation tools	Moderate
FL	Moderate to High (on non-iid data)	Low to Moderate (per node)	High (distributed training)	Ensures data privacy, reduces communication costs	Needs coordination among nodes, communication overhead	Moderate	IoT + edge devices or low-power nodes	High
IoT + Fuzzy Logic/ML	Moderate to High	Low to Moderate	High	Good for environmental control and automation	Integration with hardware can be challenging	Moderate	Sensor networks, microcontrollers	High
Drones + Multispectral Imaging + ML	High (for yield, health monitoring)	High	High (area-wise scalability)	Accurate spatial and visual analytics	High hardware cost, processing image data	High	UAVs, image processors, and data storage	Low to Moderate
YOLOv5 + Robotics (Smart Spraying)	High	High	Low to Moderate	Real-time, targeted action (weed/pest control)	Needs robotic infrastructure, costly deployment	High	Robotics + GPU + sensors	Low
Simulation + ML for Resource Allocation	Moderate to High	Moderate	High	Scenario planning, policy guidance	Requires domain knowledge and data integration	Moderate	Environmental modeling tools	Moderate
EGBM (Explainable GBM)	High	Moderate	High	High interpretability, suitable for tabular data	Limited to structured data, less suited for images	Low to Moderate	Basic ML infrastructure	High
Sensor Fusion ML (e.g., for irrigation)	High	Moderate	Moderate	Integrates multiple environmental parameters	Sensor calibration and noise handling are needed	Moderate	Multiple sensors + integration platform	Moderate
Stacked ML for Soil Prediction	High	High	Moderate	Good generalization, accurate spatial mapping	Computationally demanding	Moderate	GIS + ML environment	Moderate
Metaheuristic + Soft Computing (Resource Planning)	High (for multi-objective tasks)	High	Moderate	Handles uncertainty and conflicting objectives	Solution quality depends on tuning, not real-time	Moderate	Optimization software + datasets	Moderate

actionable insights without compromising the model's performance. This enhanced model transparency promotes interoperable agri-tech solutions and promotes digital job creation in data-informed agriculture.

2.8. Soil Health and Erosion Management

Land management depends heavily on agricultural yields, soil fertility, and soil erodibility. Further, another important term is Ecosystem resilience. In most agroecosystems, land is managed in an environmentally friendly manner, climate change adaptation is made easier, and data-driven decision-making is made simpler by ML and IoT technologies. Singha et al. (2025) used stacked ML methods to predict the Soil Quality Index of the Nile Basin and used geospatial and environmental factors to improve the model. Visual maps were constructed to track soil health, as well as to plan the region for soil conservation, and to depict the use of ML in geo-environmental modeling. Davis et al. (2023) linked soil health with agri-sustainability and ecological resilience and reinforced the significance of organic matter, nutrient cycling, and soil biodiversity, as well as integrated soil management practice. Grover et al. (2024) provided principles for sustainable soil management specifically applicable to conservation tillage, erosion control, and organic amendment and emphasized the need to educate farmers to maintain practices consistent with climate-smart agriculture objectives. Gutknecht et al. (2023) investigated cover crop adoption for soil health and climate adaptation, highlighting variability in success across regions and crops, and promoting participatory research. Cheema and Pires (2025) introduced an IoT-based soil analysis and crop recommendation system that uses smart sensors to collect real-time data on nutrients and moisture, applying ML to optimize crop choices and resource management, emphasizing the role of edge AI in precision soil management.

A Comparative Summary of ML Models for Agriculture is presented in Table 1, highlighting ML models by accuracy, cost, scalability, and suitability for smallholder farmers to aid in informed model selection.

2.9. Research gap

Despite the significant advances in ML applications in agriculture, several gaps in current research limit the scalability and economic feasibility of these technologies. One of the biggest problems is the absence of large-scale, long-term experiments that aid in enhancing the efficiency of ML in different types of agricultural setups. The majority of existing research uses small-scale testing in laboratory settings, which fail to depict the actual complexity of the agricultural setup (Priya et al., 2024). To revolutionize agriculture, additional research needs to be conducted to implement these tools in various climates, soils, and crops.

Another gap is the stringent emphasis on economic profitability. Although the majority of studies have proven the technical feasibility of ML in enhancing productivity and better utilization of resources, few consider the costs of embracing the technologies, especially for smallholder farmers in developing countries (Wanyama et al., 2024). The high upfront costs of purchasing and maintaining ML systems, coupled with the need for technical expertise, pose significant barriers to widespread adoption. Creation of cost-effective alternatives and models for business that make ML an accessible choice for the majority of farmers must be studied.

Integration of traditional farming systems and ML technologies is another field that needs to be researched. Although ML has several advantages, it also needs to go hand in hand with traditional farmers' wisdom and experience, especially in rural, remote regions where farming practices have been transmitted from generation to generation (Yousaf et al., 2023). More studies are needed on narrowing the gap between traditional and contemporary practices so that ML could support instead of substituting these practices. Lastly, safety and confidentiality of data are an issue of concern, particularly because farming data is consumed by ML models in massive volumes. Most farmers are concerned with sharing sensitive data regarding crop output or soil health because they fear related to ownership and misuse of data (Kethineni and Gera, 2023). Privacy-preserving technology, like FL and homomorphic encryption, has research in its early phase, and more research is required to secure the data of farmers while supporting the advantages of ML in agriculture (Xie et al., 2024).

3. Theoretical framework

This research employs a multidisciplinary theoretical perspective for the discussion of the integration of ML in agriculture. Implementation of ML crosses various operational, environmental, and socio-economic aspects. For developing an appreciation of ML as a force for change in general, this section leans on systems theory, theory of optimization, and theory of diffusion of innovation, sustainable development, theories of data privacy and security, theory of resilience, and technological change theory. These frameworks are synthesized to synthesize the gap between the empirical findings of Section 2 and their conceptual foundations.

3.1. System theory

Agricultural systems theory is presented in Fig. 1. Agriculture is a complex system in which soil, crops, climate, labor, and technology are interdependent subsystems. Systems theory represents these components as mutually related by feedback loops and dynamic interaction. ML models enhance the system perspective a step further by serving as smart control units that respond dynamically to variations in environmental inputs. For example, SVM application in soil erosion prediction and RF application in risk of degradation, an examples of environmental real-time data utilization to trigger adaptive actions (Cadenas et al., 2023). Further, decision tree and DL model-based applications in the detection of plant disease or in optimizing pesticide use follow the systems theory paradigm wherein ML serves as an intermediary between input sensor data and agricultural decision systems (Aqel et al., 2022).

3.2. Optimization theory

The optimization theory framework is shown in Fig. 2. Optimization theory gives a mathematical based framework for enhancing the effectiveness of resource allocation, optimizing yields, and managing labor. Optimizing algorithms like GBM, RL, and Gaussian process models are widely used to achieve this purpose. For yield prediction and crop selection, GBMs and Gaussian processes help in dealing with climate risk and crop selection of high-yielding plants according to soil and environmental factors (Islam et al., 2024). For supply chain management,

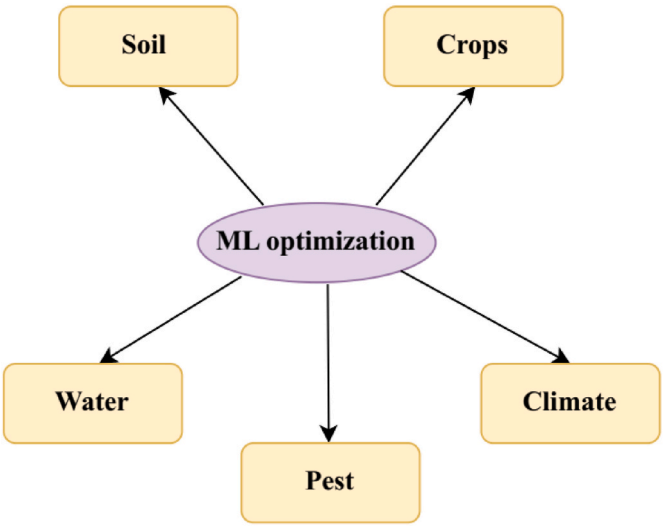


Fig. 1. System theory in agriculture.

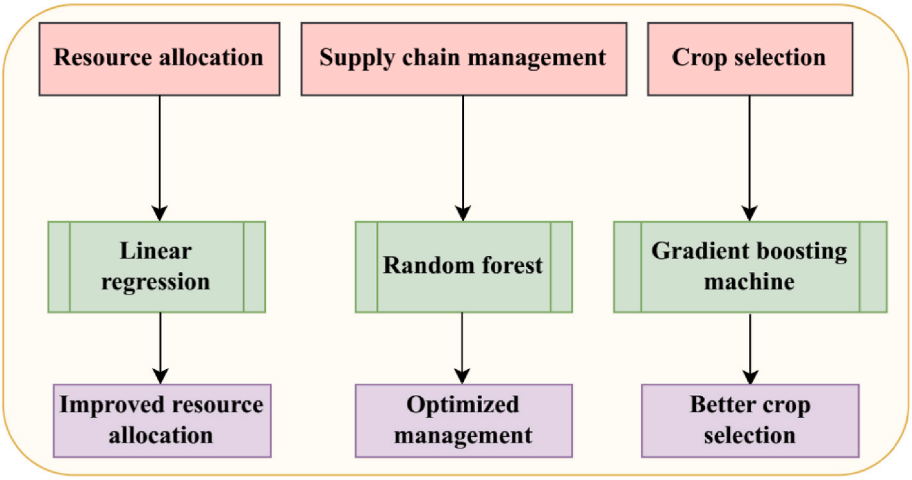


Fig. 2. Optimization theory.

SVM, RF, and Recurrent Neural Network (RNN) models have been employed in forecasting transportation delay, inventory optimization, and process streamlining (Mamoudan et al., 2023). For the sake of illustration, Fig. 2 shows a simple mapping of ML techniques to specific agricultural tasks; however, it should be noted that this is not a comprehensive representation of the breadth and complexity of ML applications in agriculture. Numerous tasks use a range of methods, such as DL models (like CNNs and LSTMs), reinforcement learning (RL), tree-based models (like RF, GBM, and XGBoost), and FL, as covered in Section 2. These tasks include crop selection, yield prediction, resource management, supply chain optimization, and environmental monitoring. These methods are not limited to a single application; for example, LSTMs are employed in both yield prediction and logistics, and reinforcement learning is used for both supply routing and irrigation scheduling.

Therefore, the purpose of Fig. 2 is not to suggest one-to-one relationships between models and tasks, but rather to provide an accessible entry point for understanding how ML is integrated into agricultural workflows. The theoretical framework outlined in Section 3 further supports this by offering conceptual foundations such as systems thinking, optimization, and resilience that explain the dynamic, interdisciplinary, and non-linear selection of ML methods in real-world agricultural applications.

3.3. *Diffusion of innovation theory*

The S-curve model of diffusion of innovation theory is presented in Fig. 3. The adoption of ML technologies in agriculture often follows the principles outlined in Rogers' Diffusion of Innovation Theory. This theory suggests that technology adoption proceeds in stages, such as innovation, early adoption, majority uptake, and laggards forming an S-curve over time. Large, tech-savvy farms are likely to be the first to adopt ML applications in precision agriculture, such as drone-based disease detection or intelligent irrigation systems, before smaller farms can access them (Rane and Choudhary, 2023). Technologies such as technological complexity, lack of trialability, and perceived incompatibility with traditionally practiced farm operations slow down adoption, particularly in developing nations (Dhanaraju et al., 2022).

3.4. *Sustainable development theory*

The ML theory of sustainable development is illustrated by Fig. 4. The theory of sustainable development focuses on balancing agricultural production with ecological conservation and social justice. ML achieves this balance by accurate and optimum use of inputs, reducing environmental degradation, and maximizing long-term sustainability. For instance, RL-based irrigation optimizes water use without varying yield

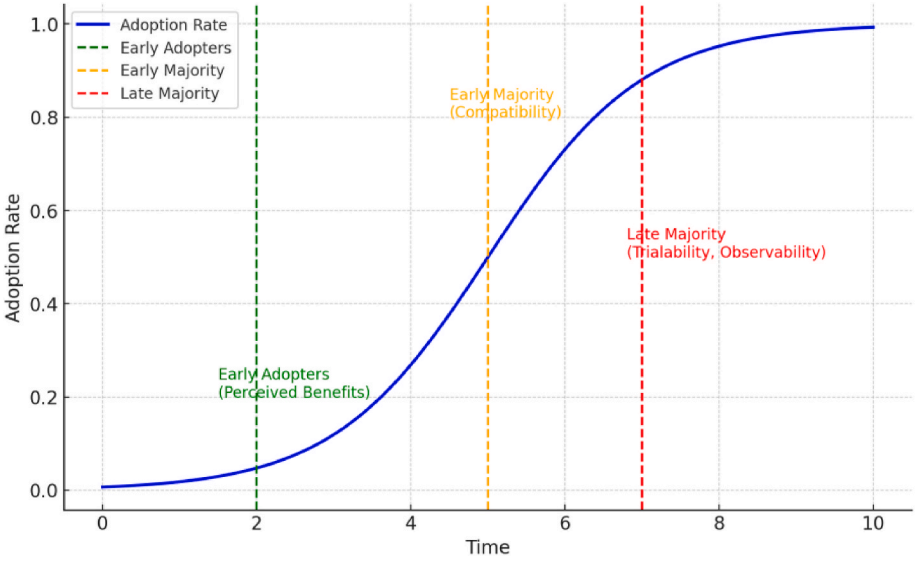


Fig. 3. S-curve model of diffusion of innovation in agriculture.

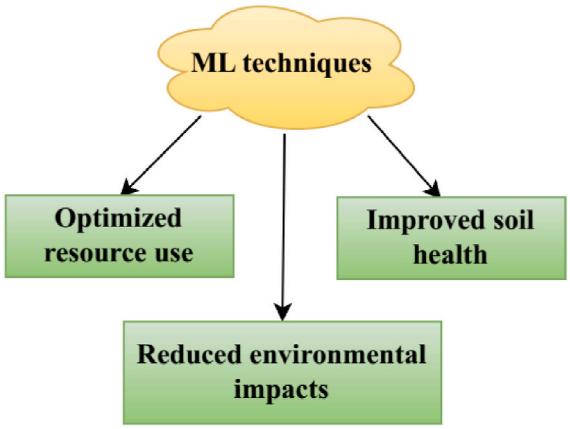


Fig. 4. ML-Driven sustainable development in agriculture.

(You et al., 2022). Analogously, ML models of fertilizer and pesticide application minimize emissions and chemical runoff (Gupta, 2023). Such applications contribute to sustainable development values by promoting wise, sustainable, and just agricultural practices.

3.5. Data privacy and security theories

Use of ML for agriculture raises privacy and ethics-related issues, particularly for confidential farm information. Privacy-preserving computation, data sovereignty, and fairness-aware learning are the principles that act as the solution to these concerns. FL ensures decentralized training of ML models without centralizing data, thereby ensuring data sovereignty. Differential privacy methods also guard individual data from inferential or abusive use (Kaur et al., 2022). Moreover, fairness-sensitive ML is also aimed at preventing biased discriminations that result from unbalanced distributions of data so that models do not disproportionately penalize marginalized or smallholder farmers (Shrikaant, 2024).

3.6. Resilience theory

Resilience theory is exemplified through Fig. 5. Resilience theory focuses on the ability of systems to withstand shocks and not become disabled when subjected to stress. ML enhances farm resilience by early recognition of weather anomalies, infestations, or supply chain disruptions. LSTM and Gaussian process-based models are best suited to predict such threats (Ghritlahare and Birali, 2024). For instance, predictive models can alert against droughts or market downturns so that farmers can undertake pre-emptive changes in planting schedules or crop diversification, and thus sustain productivity and stability under changing situations (Esteso et al., 2022).

3.7. Theory of Technological Change

Theory of Technological Change holds that innovation reshapes industries by changing production processes and labor structure. ML is at the core of such a transformation of agriculture because it allows automating the grunt work, data-driven decision-making, and interfacing with novel technologies like UAVs and IoT. The Integrative Technological Innovation Framework is employed here, which considers the co-evolution of ML with standard routines, existing infrastructure, and socioeconomic constraints (Khafagy and Vigani, 2022). For instance, the integration of autoencoders using IoT sensors is employed in job opportunities enhancement and production planning (Balogun et al., 2022). All these shifts mean that ML doesn't merely replace usual procedures but refines and puts them into practice in the new contexts of technology.

The agricultural technology theory integration into innovation validates the article, where it indicates that the intersection of the technological frontiers with the traditional farm practices leads to increased productivity, sustainability, and cost. The theory is a dynamic interaction of the technological frontiers, the traditional farm practices, and the system factors, including access to resources and environmental conditions. It points out that the successful integration of ML, to the same extent as technology, is dependent on social, economic, and environmental contexts in which it's applied. By its incorporation of the research within the theory, the research aims to investigate many ML strategies from predictive modeling and image handling to optimization techniques that are strictly utilized to address traditional agricultural issues to create a more adaptive and future sustainable agriculture. This theoretical framework provides a strong foundation for the empirical applications of ML to agriculture and spells out, explicitly, theory-to-practice links that guide subsequent development of the conceptual framework and hypotheses.

4. Conceptual framework

The application of ML in agriculture addresses critical issues like optimization of resources, sustainability, and economic viability by providing innovative solutions. Enhanced decision-making through data-led strategies reduces the environmental footprint and increases efficiency. This strategy includes the concept of ML technologies that increase the efficiency of resources employed, identify crop disease in early stages, reduce costs, and enhance resilience to climate. Through a combination of innovation and traditional processes, ML offers agriculture a dynamic and sustainable future while holding the key to balancing productivity and conservation. The conceptual framework of the current research is depicted in Fig. 6.

4.1. Resource optimization and Emission Reduction

In agricultural practices, to adopt better utilization of resources and low emissions, ML approaches are applied. The level of resources like

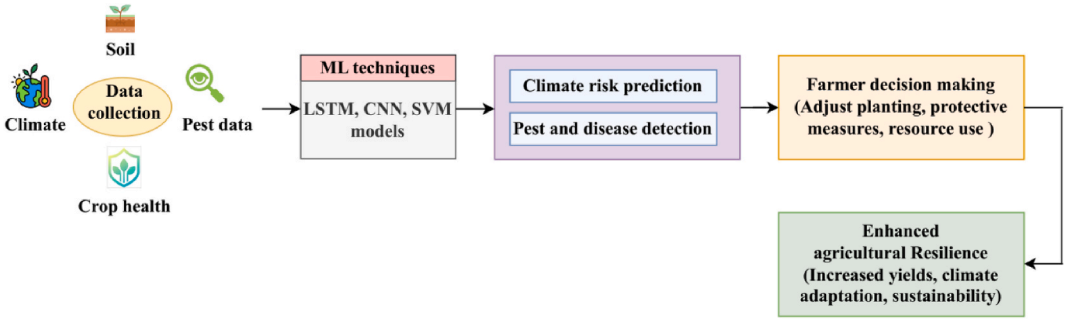


Fig. 5. Enhancing agricultural resilience with an ML approach.

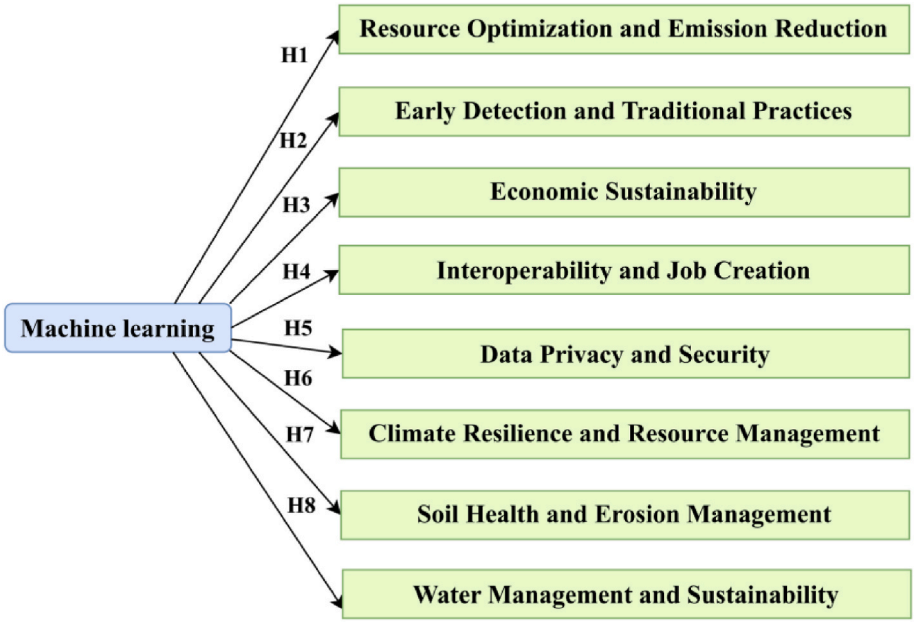


Fig. 6. Conceptual framework.

fertilizers, water, and pesticides applied during the growth of crops was estimated by utilizing ML models based on actual and historical data. For obtaining optimized irrigation and avoiding wastage of water, the level of moisture in the soil and weather are forecasted by using forecasting models (Farahzadi and Kioumarsi, 2023). Further, ML methods such as RF and SVM enable the monitoring and analysis of crop health in real-time, enabling decision-making among farmers based on resource utilization. The models not only reduce excessive usage but also cut down the utilization of greenhouse gases in agricultural activities substantially. Data-driven methods enable farmers to adopt precision agriculture in the optimum use of fertilizer and hence minimize operational expenses and optimize productivity. Through feedback loops with IoT sensors, farmers adjust practices optimally under real-time environmental conditions, with an added assurance of sustainability (Prabu et al., 2024). This hypothesis is vital for creating an agricultural framework that balances productivity with environmental protection, ultimately leading to a more sustainable future.

H1. *ML optimizes resource use, reduces emissions, and improves crop yields, profitability, and resilience across diverse agricultural climates.*

4.2. Early Detection and Traditional Practices

Apart from complementing conventional agriculture practice, ML significantly simplifies the earlier identification of diseases and plant growth disorders in crops. The farmers capture images of their crops with the use of drones or smartphones, which use advanced ML algorithms for pattern recognition and image recognition analysis. These systems process the visual information with environmental factors to detect early disease symptoms, usually at early stages (Thakur et al., 2023). This is an anticipatory measure allowing farmers to solve problems before they become catastrophic, minimize crop loss, and achieve maximum yields. Through the application of CNN and Decision Trees, ML systems are in a position to make accurate predictions of disease outbreaks. It enables farmers to maintain their local approaches, like gauging crop health based on visual inspection, and still gain from the supplementary knowledge offered by ML technologies (Nti et al., 2023). The hypothesis states that the adoption of ML in conventional policy enhances productivity in farming and cultural permanence, aimed at further entrenching the agrarian system.

H2. *ML enhances early detection of crop diseases and growth anomalies, boosting productivity and supporting traditional farming methods for smoother adoption.*

4.3. Economic sustainability

Though the application of ML technologies is extremely expensive in the early phase, ML farms are highly economically viable over time in terms of lower operation costs and efficient supply chain management. Based on cost efficiency through predictive analytics-based cost forecasting, harvest yield prediction, and market forecasting, ML algorithms can provide farmers more lucrative financial choices. ML methods optimize input utilization, which results in less input, such as fertilizers and water, with substantial long-term cost savings (Pasupuleti et al., 2024). Linear Regression (LR) and GBM methods allow farmers to estimate different sets of conditions and estimate probable monetary effects. In addition, ML increases the supply chain's efficiency through anticipating food intake behavior and optimizing supply routes, reducing losses, and maximizing the delivery time of the products. Consequently, farmers reduce operation costs while achieving maximum profitability (Javaid et al., 2023). This hypothesis highlights that the integrated approach ultimately promotes economic sustainability, demonstrating that the initial investment in ML technologies yields significant long-term financial benefits.

H3. *Despite high initial costs, ML-driven agriculture proves economically sustainable by reducing operational costs and improving supply chain efficiency.*

4.4. Interoperability and Job Creation

Interoperability among ML systems is necessary to scale and make farming more efficient and, at the same time, generate new technological employment opportunities. With the convergence of different agro platforms like IoT devices, farm management software, and analytics software, ML systems become accessible to legacy as well as contemporary farming organizations. Integration enables farmers to implement data-centric approaches, leading to improved decision-making, increased efficiency in using resources, and overall efficiency improvement (Roy, 2023). Technologies like Automated ML (AutoML) and XGBoost enable the use of sophisticated ML models in a way that is made

available to farmers who do not necessarily have sophisticated technical skills. Ease of application not only makes it easy for farmers to utilize sophisticated technology but also creates more demand for trained professionals working in data science, ML engineering, and technical support. As the agricultural sector increasingly relies on these technologies, job opportunities grow, promoting a new workforce skilled in innovative farming practices and technology (Senapaty et al., 2024). This hypothesis underscores the fact that the interoperability of ML systems and the creation of employment ensure that agriculture develops sustainably, both productively and economically.

H4. *Interoperable ML systems enhance scalability and effectiveness in agriculture, creating new tech jobs.*

4.5. Data privacy and security

A strong data protection and privacy paradigm is required for ML-based agriculture activities to safeguard farmers' data without shutting down the benefits of sophisticated technology. In conventional agricultural activities, unprotected and informal exchange of information is widespread, which results in abuse, exploitation, and mistrust. The framework suggested confronts these ethical issues with the embrace of secure, open, and farmer-driven data stewardship practices that put the farmers wholly in control of how their data is gathered, stored, exchanged, and traded. To achieve privacy while enabling collective intelligence, the platform incorporates frontier privacy-preserving technologies such as FL, Differential Privacy, and Homomorphic Encryption. FL enables edge devices to train models in a decentralized manner without exposing raw data, keeping things at home, and establishing trust for digital agri-platforms (Kaur et al., 2022). Individual identities are safeguarded by differential privacy by introducing noise while training the model, while homomorphic encryption allows computation over encrypted data without disclosing its contents and hence improving data security (Sezer et al., 2023). Ethical consideration also includes algorithmic fairness. Since ML models learn from biased or non-representative data, there is a possibility for biases while allocating disproportionate resources. For this to be avoided, the system uses fairness-aware learning and ongoing model auditing to ensure ethical decision-making, open communication, and fairness. The combined protection does not merely facilitate responsible AI adoption by the agriculture sector but also ensures a safe, inclusive, and scalable ML setting that increases productivity and digital empowerment.

H5. *Robust ML-based data privacy and security in agricultural practices lead to enhanced protection of farmer data.*

4.6. Climate Resilience and Resource Management

The addition of ML models with climate strategies helps to improve the resilience of agriculture to climate variability and improve resource management practices. ML algorithms can identify climate patterns and forecast climate-related impacts to agriculture with the help of historical climate data, remote-sensing data, and other data. Access to information empowers farmers to modify strategies (Benitez-Alfonso et al., 2023). The KNN model uses historical yields of crops and soil data to recommend potential crop rotation plans (Resti et al., 2022). LSTM Networks and Gaussian Process Regression models allow farmers to predict climate-related risks, even without real-time sensor data. These models take existing data to highlight correlations and trends in the available data to make decisions surrounding resource allocation, such as irrigation scheduling, possible crops to grow, etc. Using ML strategies, farmers enhance resilience to climate variability and improve the efficiency of resource usage (Kumar et al., 2022). This hypothesis is instrumental in developing adaptive strategies to ensure stable production in climate change, promoting long-term agricultural sustainability.

H6. *ML models integrated with climate strategies improve agriculture's resilience to climate variability and enhance resource management practices.*

4.7. Soil Health and Erosion Management

ML enables farmers to predict and manage soil erosion risk, therefore improving soil health and decreasing land degradation. By analyzing key soil data variables, moisture levels measured by soil sensors, ML models provide the risk associated with soil degradation and erosion. The models process real-time data collected from soil sensors that measure soil moisture, nutrients, and other environmental conditions for an understanding of the soil health (Mohanty et al., 2024). Farmers can take a proactive approach to ensure soil health by using ML predictions to amend practices, such as irrigation changes or soil inputs, to bolster nutrient levels and to decrease soil erosion. For this predictive activity, farmers could also use models like Support Vector Regression (SVR) and RF Regressor to evaluate all these measures and suggest improvements to soil health to lessen soil erosion. By using ML with sensor data to preserve, improve, and restore sustainability, farmers can sustainably manage their land (Silva et al., 2024). This hypothesis highlights the importance of maintaining soil fertility, which not only reduces the need for chemical inputs but also safeguards land resources for future agricultural use, ensuring long-term sustainability.

H7. *ML predicts and manages soil erosion risks, contributing to improved soil health and reducing land degradation.*

4.8. Water management and sustainability

Water savings and sustainable agricultural methods are critically important for agricultural production and irrigation management because ML intends to predict water needs for crops. The farmer can modify irrigation schedules using predictions based on ML models, with the understanding that this will minimize the loss of water and improve agricultural output. KNN and ANN are generally the two most used ML techniques that help farmers process a range of input data: soil moisture sensors, weather variables (temperature, humidity, and precipitation), and past irrigation uses (Czekała and Pulka, 2024). The first step is for farmers to fill data sources from the soil sensors and weather sensors, whereby the farmer collects and pre-processes the data and develops to ensure the data and information are accurate and relevant. The farmer then undertakes the ML training on the data so they can learn the patterns of relationships between the environmental and water needs of the crops. Ultimately, the trained models are put together by the developers for the farmers to develop their best combinations of irrigation timing and volumes, and the farmers can manipulate them to suit their timing and watering schedule. In addition to these techniques, Dynamic Bayesian Networks (DBNs) have emerged as a robust method, particularly when sample sizes are limited or when managing uncertainty in temporal data (Ma et al., 2024a,b). This demonstrated the effectiveness of DBNs in assessing crop water productivity and optimizing planting structures in arid irrigation districts. DBNs can integrate probabilistic reasoning and time-dependent variables, making them particularly suitable for modeling water usage over time under varying environmental conditions. By incorporating ML methods, including DBNs, farmers can develop adaptive and data-driven irrigation systems. This hypothesis underscores the critical role of sustainable water management in agriculture, contributing to improved crop yields and responsible water usage, both of which are essential for long-term agricultural viability.

H8. *ML-driven irrigation management reduces water consumption, promoting sustainable practices and enhancing agricultural productivity.*

The use of ML in agriculture offers transformative resolutions to serious problems such as resource allocation, sustainability, economic viability, and environmental security. Each hypothesis in this

framework articulates specific ML solutions, including resource use, climate resilience, data security, and early disease identification, and shows ways farmers can achieve greater productivity, reduced costs of operations, and promote environmental sustainability. This framework articulates how ML-driven agriculture is using more sophisticated models such as RF, CNN, and AutoML, and not only is increasing economic growth, but also creates tech jobs and gives digital technology access to farmers. Therefore, ML can be a key driver of economic growth, creating and maintaining a sustainable, resilient, and secure agricultural landscape, ensuring that the industry evolves to meet modern demands while safeguarding both ecological integrity and financial stability.

5. Methodology

The purpose of this paper is to evaluate the applications of ML techniques in agriculture. The review takes a comprehensive approach in critically analyzing all aspects of data gathering, data analysis, and data validation in the research context to provide reliable findings. The paper carries out a defined quantitative research design. The methodology used provides research that is reliable, transparent, and follows the principles of rigor and completeness. This section describes the research design, the target population and sampling process, data collection procedures, and the statistical analysis used.

5.1. Research design

A cross-sectional research design was employed in this study, as it was the most suitable approach for analyzing the adoption of ML in agriculture. The research was purely based on quantitative data and offered the opportunity to draw general conclusions based on empirical evidence. The literature review was completed by searching reputable academic and scientific online databases, such as IEEE Xplore (https://ieeexplore.ieee.org/Xplore/home.jsp), Scopus (https://www.scopus.com/), and Google Scholar (https://scholar.google.com/). These were not only the most widespread databases for disciplines such as engineering and computer science, but also for agriculture. This also ensured that the study included a wide cross-section of academic research relevant to ML and agriculture. These key terms were employed to search the databases: "machine learning" AND "sustainable agriculture" AND "precision agriculture". These keywords allowed an in-depth exploration of key areas in which ML is influencing agricultural practices, such as crop disease detection, resource optimization, climate resiliency, and economic resiliency. To maintain focus and ensure the literature was relevant, the following inclusion and exclusion criteria were applied: only peer-reviewed journal articles and conference papers published between 2022 and 2025 were considered eligible, and the application of ML specifically for agricultural situations was focused on. Eligible studies encompassed empirical research, review articles, and theoretical papers providing insights into the impact of ML on agriculture, and the articles had to be published in English. Excluded were papers discussing ML without any agricultural application, non-peer-reviewed publications (such as reports, blogs, or opinion pieces), studies lacking significant empirical evidence or analysis of ML's role in agriculture, and duplicate studies from multiple databases. The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram is depicted in Fig. 7.

5.2. Study area

The study area of the proposed research is illustrated in Fig. 8. It encompasses three key agricultural regions in China, such as Jiangsu, Shandong, and Xinjiang, each selected for its distinct and dominant farming practice. Jiangsu Province is widely recognized for its intensive rice farming, Shandong Province is a national leader in vegetable cultivation, and Xinjiang Province is known for its expansive fruit production, particularly melons and grapes. These regions were purposefully selected not only because of their agricultural significance and crop diversity, but also due to their varying climatic conditions, economic

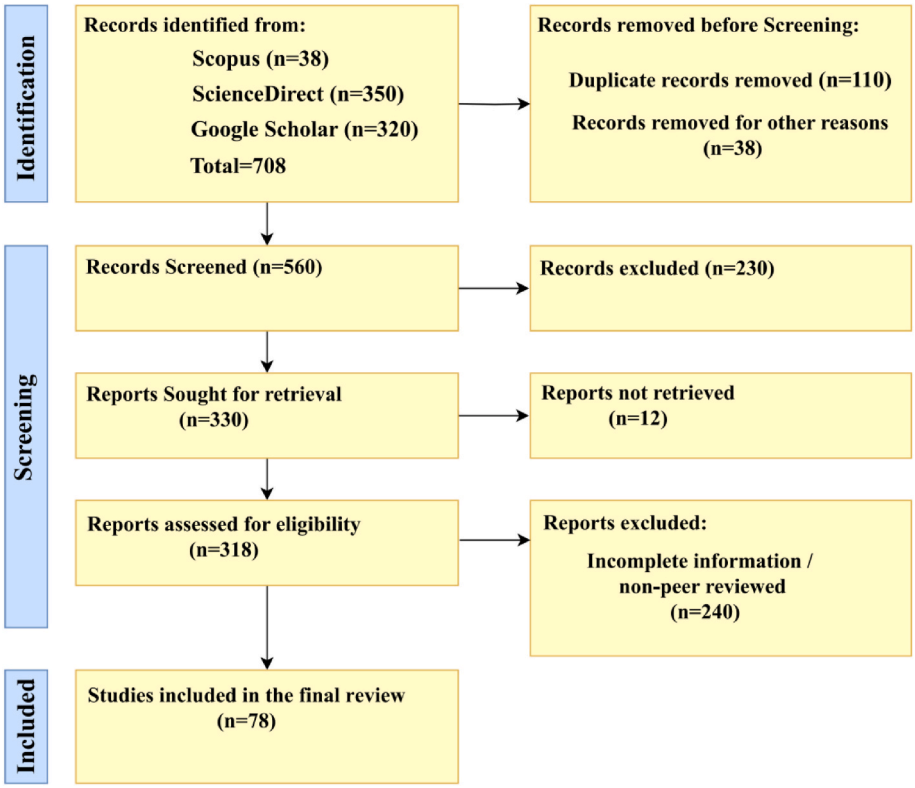


Fig. 7. PRISMA flow diagram outlining the systematic selection of studies. Records were sourced from Scopus, ScienceDirect, and Google Scholar. After duplicate removal and screening, 78 studies were included in the final review.

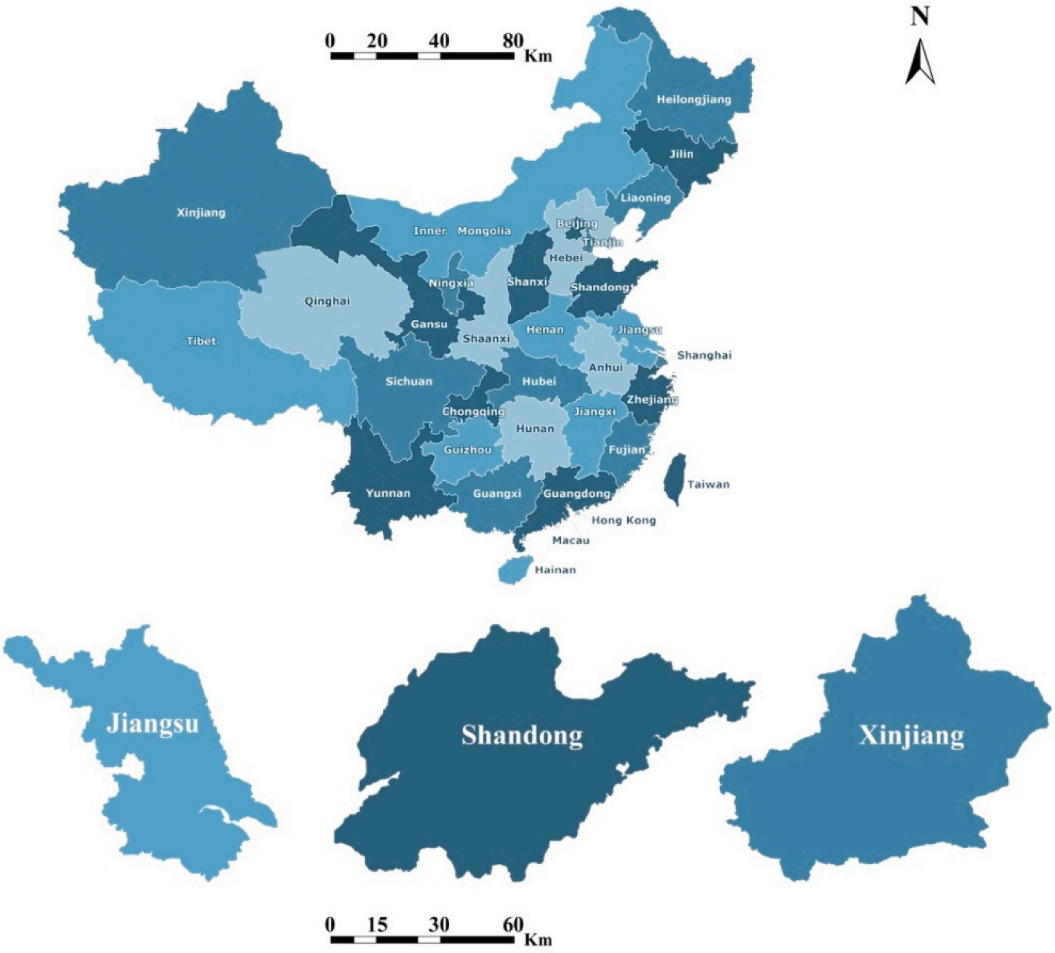


Fig. 8. Study area.

development levels, and technological readiness, which offer a comprehensive cross-section of China's agricultural landscape. Furthermore, farmers in these regions have demonstrated substantial experience with the adoption of modern agricultural technologies, including ML applications, making them ideal for evaluating the feasibility and challenges of ML-based interventions in real-world agricultural settings. Although the study is focused on China, the insights gained, especially in terms of adoption barriers, farmer perceptions, and effectiveness of ML tools, have broader relevance for other countries with similar agricultural profiles, particularly in emerging economies facing challenges related to climate change, labor shortages, and precision farming adoption. Therefore, while grounded in a local context, the study findings contribute to a global understanding of ML integration in agriculture, especially in regions where crop specialization and digital transformation are increasingly prioritized.

5.3. Target population

The target population of this study comprises experienced farmers, individuals with agricultural knowledge, agricultural-related personnel, and farmers without technical knowledge. Individuals with agricultural knowledge are the person who may not actively farm but have formal education or training in agricultural sciences, agronomy, or related fields. The theoretical perspectives of the tools within the agricultural technologies that include sensors, drones, etc, are provided with the understanding of ML incorporation to current practices. The professionals, such as agricultural extension workers, consultants, and researchers, come under the agricultural-related personnel category. These professionals aid farmers in developing novel technologies and best practices. The efficiency as well as practicality of the ML applications in agriculture is assessed with their expertise. In circumstances involving farmers with very little to no experience with modern technologies (i.e., farmers do not have a technical background), progress should be made in understanding usability, accessibility, and training of the new program's ML technologies, making ML easier to incorporate for all farmers, regardless of their technical background. By developing a diversified target population, the study aims to capture a comprehensive view of the potential impact of ML technologies on agricultural practices, which includes the pros and cons of using ML technologies with different levels of experience and expertise.

5.4. Data collection

A survey questionnaire was distributed to assess the experiences and attitudes of 847 participants regarding ML technologies in agriculture. When the data were collected, they were meticulously cleaned to eliminate any incomplete or inappropriate responses, resulting in 793 responses. From this sample, participants represent a broad range of experiences related to the viability, limits, and success of ML adoption in everyday agricultural practices. Statistical Package for the Social Sciences (SPSS) was then used for data analysis after the completion of the cleaning. The questionnaire was designed with both closed and open questions, where participants shared information about their opinions and perceptions of ML techniques. The survey raised a variety of things related to ML in agriculture, including the participants' awareness of ML applications in agriculture; participants' self-reported views of their

receptiveness to new technology; perceived limitations to ML implementation, e.g., costs, knowledge, etc; sources of information for agricultural technologies; and interest in support and training for adoption.

5.4.1. Questionnaire design and validation

The questionnaire was designed and validated from a review of current literature on technology acceptance in agriculture and the use of (previously validated) survey designs, which were used in similar studies. The questions were designed to align with the study's objectives, particularly in understanding awareness, adoption barriers, and motivations related to ML in farming practices. To ensure content validity, the initial draft of the questionnaire was reviewed by a panel of three experts in agricultural technology and survey methodology. A pilot test was conducted with 10 participants from the target population to refine the language, clarity, and structure of the items. Feedback from the pilot led to minor adjustments in wording to improve comprehension.

5.4.2. Internal model evaluation and technical validation

While model performance was evaluated through a structured survey due to the management research context, we also conducted internal performance checks on each ML model to ensure functional validity. For instance, CNNs were evaluated on image datasets using accuracy and precision to confirm effective feature extraction, and LSTMs were tested on time-series crop yield data using RMSE for trend forecasting. Hyperparameters were tuned using grid search and 5-fold cross-validation on limited datasets. Preprocessing steps included normalization, handling missing values, and data augmentation for image-based tasks. Training duration varied depending on model complexity and dataset size. For instance, LSTM models were trained for up to 50 epochs on time-series data, while CNN models for image classification tasks were trained for 30 epochs on augmented samples. Though these technical metrics were not the focus, they ensured each model performed sufficiently before interpreting results through survey outcomes.

5.5. Quantitative analysis of reviewed literature

The questionnaire survey is presented in Table 2.

Table 3 provides the demographic profile of the participants. Based on gender, 56.66 % were male and 43.33 % were female. As per the age of the participants, 10 % of farmers were aged up to 30 (Young), 23.33 % of farmers aged 31–50 (Middle), and 66.66 % of farmers aged above 50 (old). In terms of education, 66.66 % of the participants had completed up to 6 years of formal schooling (primary level), 26.66 % had completed between 7 and 12 years of schooling (secondary level), and 6.66 % had completed more than 12 years of education (higher secondary or above). Based on the main occupation of the participants, 70 % of participants were Rice, vegetable, and fruit farmers, 16.66 % of participants were working in field crops, 11.66 % of participants were merchants, and 1.66 % of participants were working off-farm. Following the experience of rice, fruit, and vegetable farming, 20 % of farmers have experience up to 20 years, 70 % of farmers have experience 21–40 years, and 10 % of farmers have experience above 41 years. As per the total land area, 20 % of farmers have land areas up to 5 acres, 66.66 % of farmers have land areas of 6 to 5 acres, 10 % farmers have land areas of 16–25 acres, and 3.33 % of farmers have land areas of more than 26 acres. Based on the land tenure, 76.66 % of the participants are owners of their land, 20 % of the participants' land is rented land, and 3.33 % of the participants have the land for free. As per the title to the land, 8.33 % of participants' lands are certificates of utilization, 6.66 % of participants' lands are agricultural land reform, 3.33 % of participants' land has no evidence, and 1.66 % of participants' lands are not specified.

Table 2
Questionnaire survey.

Question	Response Options
1. Are you aware of ML applications?	A. Yes/B. No/C. Unsure
2. Interest in learning new technologies?	A. Very interested/B. Somewhat interested/C. Not interested
3. Perceived barriers to ML adoption?	A. Cost/B. Lack of knowledge/C. Lack of access/D. Other
4. What kind of support would help?	A. Training/B. Financial assistance/C. Access to info
5. Use of technology in farming?	A. Yes/B. No
6. How often do you seek information?	A. Never/B. Rarely/C. Sometimes/D. Often
7. Sources of information on agriculture?	A. Family/Friends/B. Workshops/C. Online resources
8. Motivation for adopting technologies?	A. Increased productivity/B. Financial incentives/C. Environmental benefits

Table 3
Demographic details of farmers.

Variables	Categories	Frequency	Percentage
Gender	Male	170	56.66
	Female	130	43.33
Age	Young (up to 30)	30	10
	Middle (31–50)	70	23.33
	Old (>50)	200	66.66
Education Qualification	Up to 6 years of schooling	200	66.66
	7–12 years of schooling	80	26.66
	More than 12 years of schooling	20	6.66
Main Occupation	Farmer (rice, vegetable, fruit)	210	70
	Field crops	50	16.66
	Merchant	35	11.66
	Off-farm	5	1.66
Farming Experience	Up to 20 years	60	20
	21–40 years	210	70
	>41 years	30	10
Total land area	Up to 5 acres	60	20
	6–15 acres	200	66.66
	16–25 acres	30	10
	>26 acres	10	3.33
Land tenure	Owner	230	76.66
	Rental	60	20
	Free	10	3.33
Title to Land	Title deed	240	80
	Certificate of utilization	25	8.33
	Agricultural land reform	20	6.66
	No evidence	10	3.33
	Not specify	5	1.66

5.6. Statistical analysis

The data collected from the questionnaire were analyzed using **SPSS** software. The following statistical techniques were employed to test the study's hypotheses and analyze trends.

- ***Descriptive Statistics:*** Used to summarize the data and provide an overall view of participant responses.
- ***Reliability Analysis:*** Cronbach's alpha was calculated to ensure the reliability of the questionnaire responses.
- ***Hypothesis Testing:*** Regression analysis, t-tests, and correlation analysis were performed to test hypotheses related to ML adoption, resource optimization, disease detection, economic sustainability, etc.

6. Results

The paper examines the empirical examination of the utilization of ML techniques in the agricultural field. The investigation explores aspects related to reliability and validity analysis, discriminant validity analysis, and hypothesis testing employing ML techniques in

agriculture. The outcomes of these analyses are presented in the subsequent subsections, accompanied by detailed explanations.

6.1. Reliability and validity analysis

Table 4 provides the results of reliability and convergent validity tests for various variables related to the application of ML techniques in agriculture. The reliability of a variable measures the consistency and stability of the measurements. It is assessed using Cronbach's alpha (α) value, Composite Reliability (CR), and rho-A value. Higher values indicate greater reliability. The values should be greater than 0.70 to be acceptable for determining reliability. Mean (M) represents the average value for each variable, which should be greater than 3.00, and Standard Deviation (SD) indicates the degree of variation in the data. Average Variance Extracted (AVE) measures the proportion of variance captured by a construct over the variance due to measurement error; it should be more than 0.50 to achieve validity.

The reliability and convergent validity of the variables related to ML applications in agriculture were rigorously assessed to establish their credibility and effectiveness within the conceptual framework. The variable for **'Resource Optimization and Emission Reduction'** demonstrates high reliability (α = 0.860) and good convergent validity (AVE = 0.671). This means that it is a strong assessment, obviously to reduce key losses of resources and greenhouse gases, and to support environmental sustainability. The reliability measures indicate that ML algorithms provide good predictive ability to allow remediation on time to manage resources. Concerning 'Early Detection and Traditional Practices,' this variable has very reliable (α = 0.955) and good convergent validity (AVE = 0.722). The results, therefore, demonstrate that ML approaches to precision agriculture, such as improving crop management practices or improving traditional practices, are effective in improving agricultural outcomes. The agriculture field plays a pivotal role in food production, and therefore validating the harvesting of those practices for ML application to maintain food security is vital. The variable 'Economic Sustainability' also has high reliability (α = 0.861) and good convergent validity (AVE = 0.591). ML applications help improve economic still in agriculture when they can interrogate economic data and provide recommendations for improving costs and maximizing profits. The dependability of this variable is indicative of its indicators in increasing sustainable economic practices in the agricultural domain. The variable 'Interoperability and Job Creation' demonstrated relatively good reliability (α = 0.800) and convergent validity (AVE = 0.626). ML applications help create interoperable systems of agricultural operations that efficiently utilize resources and, therefore, create more jobs in agriculture. Efficiency in resource utilization leads to more profit for farmers while supporting sustainable practices by minimizing waste, which is why this measure is a vital aspect of agricultural economics.

The variable, 'Data Privacy and Security', has acceptable reliability (α = 0.812), and good, but not great, convergent validity (AVE = 0.789); this means that ML technologies used for protecting sensitive agricultural data to a level of competence. By supporting opportunities for data privacy, ML maintains farmer trustworthiness and motivates them to adopt technological innovations in agriculture. The variable, 'Climate Resilience and Resource Management', has moderate reliability (α = 0.815), and only moderate, though officially acceptable, convergent validity (AVE = 0.666) as well. In the case of climate resilience, ML applications are the only choice for adapting agricultural practices to changing environments, and they ensure that farmers can do so while effectively managing their resource allocation to reduce climate risks. With change being a massive commodity risk in agriculture, it is important to effectively measure climate resilience and support sustainable agricultural practices. The variable 'Soil Health and Erosion Management' reveals high levels of reliability (α = 0.834) with a pretty good (but not great) level of convergent validity (AVE = 0.653). The applications of ML are key factors for monitoring soil health and for developing and implementing erosion prevention practices, since they speak to food productivity and sustainability in agriculture. The reliability of this variable illustrates the necessity for advanced strategies involving technologies in soil management practices. The final variable, 'Water Management and Sustainability,' had an excellent reliability value (α = 0.905) and convergent validity (AVE = 0.700). ML applications can assist with water management of crops and ensure appropriate irrigation practices and moisture levels for plants, all while reducing the precious resource of water use. Water scarcity in farming and growing food is a growing problem, and potentially the most important measure of sustainable cultivation. Overall, the evaluations of the distinguishing variables associated with ML applications in agriculture measure reliability and good convergent validity in sustainability and productivity in agriculture.

6.2. Discriminant validity analysis

Under convergent validity using AVE and Cronbach's alpha (α) met their threshold values to illustrate validity. The results for discriminant validity, measuring the divergence of each variable in the study, are displayed in Table 5. It presents a matrix of squared correlations between the variables. Each cell in the matrix represents the squared correlation coefficient between two variables. Confirming that each variable exhibits a higher variance correlation with its measures than with measures of other variables (as shown in italics in Table 5) has solidified the discriminant validity of the examined measures.

To confirm discriminant validity, each variable must demonstrate a higher variance correlation with its measures than with those of other variables, as evidenced in Table 5. The Resource Optimization and Emission Reduction variable exhibits a squared correlation of 0.819, reflecting strong internal coherence and indicating that its measures effectively capture the essence of optimizing resources while minimizing emissions. The Early Detection and Traditional Practices variable has a squared correlation of 0.849, affirming its reliability and highlighting the importance of early detection in agriculture. Similarly, the Soil Health and Erosion Management variable had a square correlation of 0.808, which provides good validity for the protection of soil health. Next, the Economic Sustainability variable had a square correlation of 0.768, demonstrating suitability for agricultural sustainability, which is important to survival long term. From here, the Interoperability and Job

Table 4
Reliability and Convergent validity test results.

Variables	Reliability			Descriptive Statistics		Convergent validity	Result
	α-value	CR	rho-A	Mean (M)	SD	AVE	
Resource Optimization and Emission Reduction	0.860	0.885	0.861	3.62	0.731	0.671	Valid & Reliable
Early Detection and Traditional Practices	0.955	0.961	0.959	4.50	0.545	0.722	Valid & Reliable
Economic Sustainability	0.861	0.882	0.865	3.52	0.670	0.591	Valid & Reliable
Interoperability and Job Creation	0.800	0.823	0.810	3.43	0.956	0.626	Valid & Reliable
Data Privacy and Security	0.812	0.823	0.791	3.21	0.871	0.789	Valid & Reliable
Climate Resilience and Resource Management	0.815	0.857	0.820	3.20	1.123	0.666	Valid & Reliable
Soil Health and Erosion Management	0.834	0.863	0842	3.37	1.043	0.653	Valid & Reliable
Water Management and Sustainability	0.905	0.920	0.910	4.00	0.579	0.700	Valid & Reliable

Table 5
Squared Correlation and Discriminant Validity analysis.

Variables	1	2	3	4	5	6	7	8
Resource Optimization and Emission Reduction	*0.819*							
Early Detection and Traditional Practices	0.810	*0.849*						
Soil Health and Erosion Management	0.674	0.673	*0.808*					
Economic Sustainability	0.562	0.439	0.682	*0.768*				
Interoperability and Job Creation	0.634	0.542	0.430	0.342	*0.791*			
Data Privacy and Security	0.561	0.321	0.436	0.451	0.561	*0.888*		
Climate Resilience and Resource Management	0.629	0.663	0.555	0.468	0.453	0.675	*0.816*	
Water Management and Sustainability	0.700	0.725	0.610	0.580	0.670	0.690	0.750	*0.836*

Creation variable had a square correlation of 0.791, linking technology compatibility with job creation. Equally important was the Data Privacy and Security variable, which had a square correlation of 0.888, emphasizing the necessity to protect sensitive farm data. Not surprisingly, the Climate Resilience and Resource Management variable had the highest square correlation at 0.816, which illustrates the potential of ML methods for determining high-risk areas for resource management and resilience. Finally, the Water Management and Sustainability variable had a square correlation of 0.836, emphasizing that the correct strategies for water management are important. Overall, the discriminant validity analysis provides excellent evidence for the measured validity and distinctiveness of each of these variables to ensure clear operational definitions and significance to agricultural research and practice.

6.3. ML techniques in agriculture

ML techniques in agriculture refer to the application of various ML algorithms and data-driven approaches to address challenges and optimize processes within the agricultural sector. In the survey, participants were asked to provide their perspectives on various aspects of the application of ML in agriculture. 90 % of the participating farmers agreed with the notion that machine-learning techniques contribute to a reduction in overall resource consumption. Meanwhile, 7 % of respondents held a neutral perspective, and 3 % disagreed with this statement. Among the survey participants, none expressed the belief that ML is incapable of minimizing soil erosion and land degradation. Instead, 80 % of respondents agreed that ML holds the potential to address these issues, while the remaining 20 % adopted a neutral stance on the matter.

Reducing greenhouse gas emissions is a vital environmental goal. In the survey, 60 % of respondents agreed that ML techniques contribute to achieving this objective, while 36 % remained neutral on the topic. Only 4 % of participants disagreed with this notion. The majority, 89 %, agreed that ML techniques are effective in identifying crop diseases, highlighting the potential of these technologies in enhancing crop health management. When it came to the early detection of growth anomalies, 75 % of respondents agreed, indicating that ML holds promise in improving the timely identification of issues affecting crop development. Additionally, 55 % of participants acknowledged the potential of ML to advance plant breeding practices, although a sizable 40 % remained neutral on this aspect. Furthermore, in terms of crop yield production, 60 % of survey participants agreed with the potential benefits of ML, while 30 % took a neutral position, and 10 % expressed disagreement. Fig. 9 illustrates farmers' perception of the application of ML in agriculture. Overall, the responses collectively represent the respective views around the use of ML techniques within the agricultural domain as reported by individuals represented in the survey.

6.4. Traditional farming and smart farming

Traditional agriculture depends on taking action and the traditions of farming. Smart farming refers to the use of technology, data, and automation of agricultural processes to optimize farming, decrease

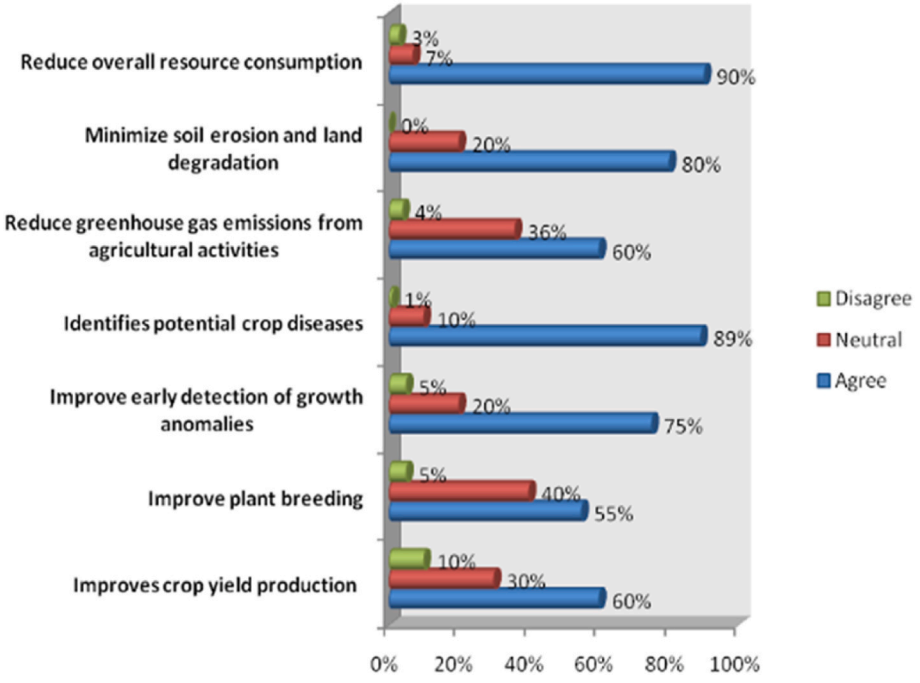


Fig. 9. Farmers' perception of the application of ML techniques in agriculture.

resource consumption, make production more sustainable, etc. As part of the survey participants' view about food production, "smart" agriculture and "traditional" agriculture were assessed based on agriculture-related tasks/tasks related to food production.

A comparison of traditional and smart farming systems is summarized in Fig. 10. The results of the survey indicated 55 % of individuals agreed that smart farming had or could contribute to enhancing the quality and yields of food produced, a reflection of the perceived benefits of technology in enhancing agricultural outcomes in terms of the quality of food produced/harvested. In addition, 75 % of individuals agreed that smart agriculture uses more resources than traditional agriculture, a consistent view that modern agriculture, driven by technology, is resource consumptive and intensive. Additionally, 70 % of participants agreed that smart agriculture enhanced productivity, another perception that technology-driven perspectives improve total agricultural efficiency. Finally, in the comparisons related to reducing waste, 60 % of the individuals agreed that smart agriculture has, or could lead to, less waste and agreed with a similar response to comparative traditional agriculture methods.

The cost-benefit and environmental analysis is presented in Table 6. This table offers a quantitative and comparative overview of traditional farming and smart farming with ML, evaluating both economic and environmental metrics, such as technology cost, maintenance, yield, resource consumption, emissions, and labor. From an environmental standpoint, smart farming demonstrates notable improvements measured through key indicators such as water footprint, carbon footprint, and lifecycle assessments (LCA). The water footprint represents the volume of water consumed during agricultural processes. Smart farming, powered by ML algorithms, reduces water usage by approximately 15 % through precision irrigation based on real-time weather data and soil moisture levels. The 15 % reduction in water usage translates to a saving of approximately 180 cubic meters per hectare per year. Assuming a local irrigation cost of $0.50 per cubic meter, this equates to $90 saved per hectare annually on water alone. The carbon footprint, which is a total measure of greenhouse gas emissions, is considerably lower for smart farming, which is caused by improved energy efficiency and lower fuel use, as well as reduced fertilizer and pesticide use through automated systems. Along with lower fuel use, the $5000 per farm reduction (which is about $250 per hectare for a 20-ha farm) in pesticide use with ML-driven targeted application represents a significant drop and directly impacts the bottom line benefit. The life cycle assessment (LCA) presents a big picture assessment of the environmental benefits of smart farming across the life cycle from production to disposal (including increases in greenhouse gas emissions

Table 6
Cost-benefit and environmental analysis of traditional vs. smart farming (with ML).

Cost/Benefit Aspect	Traditional Farming	Smart Farming (with ML)	Impact of ML
Initial Technology Cost	$2000–$3000 per farm setup	$10,000 per farm setup	Smart farming requires a higher upfront investment due to sensors, software, and training.
Annual Maintenance Cost	$500–$1000 per year	$2000 per year	Traditional maintenance costs involve basic repairs and labor, while ML systems require software updates and professional services.
Crop Yield Increase	Baseline (no significant increase)	20 % increase	Traditional farming typically produces stable yields, whereas ML optimization boosts yield by 20 %.
Water Consumption	1200 cubic meters per hectare/year	1020 cubic meters per hectare/year	Traditional farming uses more water, while ML-based precision irrigation saves about 15 %.
Pesticide Usage	$7000 per year	$2000 per year	Traditional methods involve widespread pesticide use, while ML targets problem areas, saving $5000/year.
Greenhouse Gas Emissions	15 tons CO_2 equivalent/ year	5 tons CO_2 equivalent/ year	Traditional methods contribute to more emissions due to inefficient machinery and input usage.
Profitability Increase	Baseline (no significant increase)	$12,000 increase/year	ML-driven techniques improve profitability with higher yields and better resource management.
Labor Cost	$20,000 per year	$18,000 per year	Traditional farming requires more manual labor, while ML reduces labor needs by 10 %.
Return on Investment (ROI)	N/A	2.5 years	ML investments typically pay off within 2.5 years, considering the savings and revenue increases.
Lifecycle Assessment	High	Low	Reduced resource extraction and waste due to precision farming

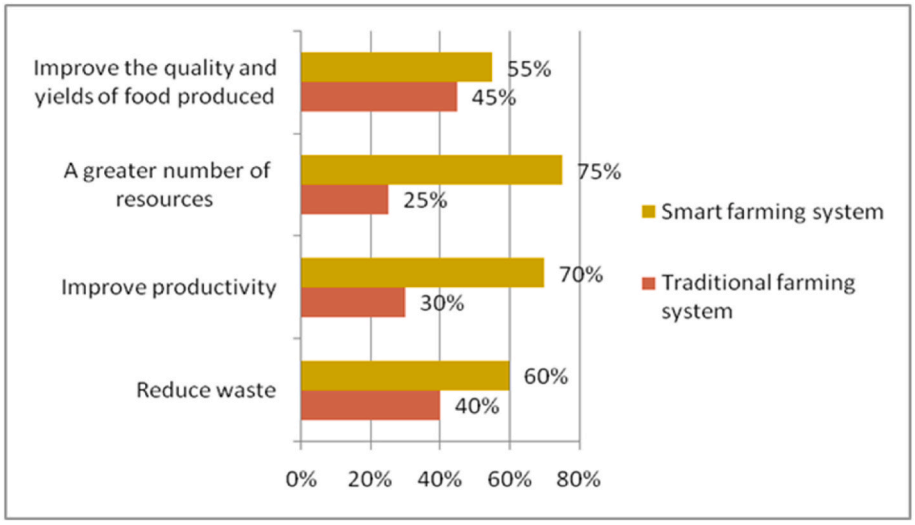


Fig. 10. Comparison of traditional and smart farming systems.

through resource extraction) and suggests that smart farming leads to less resource extraction and waste compared to traditional farming. Financially, there appears to be much more of an upfront cost (initial set-up costs of $10,000 for smart farming compared to $2000–$3000 for traditional farming), but significant ongoing annual savings and profit opportunities are substantial. From a financial perspective, these savings translate to about $12,000/year increase in profit based on a 20 % increase in yield, $5000/year savings from reduced pesticide use through targeted applications, 15 % reduction in water use, and $2000/year decrease in labour costs directly associated with automation of a variety of farming tasks. Even considering an increased annual cost of maintenance ($2000) for smart farming beyond that of traditional farming ($500–$1000), the overall net gain is significant. With this analysis, the ROI for smart farming is approximately 2.5 years and is associated with ML on demand. After this point, farmers always gain additional revenue and lessen operational costs. These are benefits each year, demonstrating that smart farming is a sustainable investment choice. Farmers can also start small with new modular systems like ML-based irrigation controllers or pest detection systems, and gradually adopt smart farming tools without replacing all their growing experience and systems. Traditional farming may seem like a less expensive option in the first year, but then higher costs stack up over time, resource misuse piles up, and yields plateau. After extensive, detailed calculations of costs and benefits and environmental aspects, the truth is clear. Traditional farming may start with fewer upfront expenses, but it incurs more long-term environmental and resource costs, and uses more people to do the same amount of overall work. ML-driven smart farming has more economic and environmental advantages and ultimately is more profitable, efficient, and sustainable. The economic benefits were presented, as were suggestions for environmentally sustainable practices over time. Therefore, it would be better for farmers to adopt ML technologies widely in a sustainability discussion, rather than say few listed above.

6.5. Comparative analysis with other data-driven methods

Although smart farming afforded through ML technology has shown strong improvements in efficiency, precision, and sustainable farming practices, it is necessary to frame any performance of its ML model against other conventional data-driven modeling methodologies (e.g., rule-based systems, statistical models (e.g., LR, Autoregressive Integrated Moving Average (ARIMA)). Given that ML-based smart farming occupies a key element of contemporary precision agriculture, it has most certainly advanced beyond previous forms of data-driven modeling presented by expert-influenced rule-based systems and levels of computerized decision-making. Any rule-based system is derived from a set of rules defined by an expert, e.g., "an irrigation system is programmed to irrigate crops if soil moisture <30 % and temperature >28 °C." As presented, rule-based systems written in terminology like above comprise interactions of high degree of familiarity, i.e., soils are diurnally stable and understood, archetypes for seeds exist in an expert's mind for plant stage requirements to thrive and develop towards their respective harvest for example; hence these rule-based systems works in static or stable fiducial environments; however, the system does not know the dynamics considered when rules are generated, nor does it account for unrecognizable states of being in a particular environment. The same premise applies to statistical modeling systems like LR and ARIMA models for reasonable predictions of multi-year crop yield trends, effective rainfall amount predictions, or barriers to crop health, such as disease or pest infection. Providing an easy-to-read document for an audience to understand with a reasonable word or numbers where computation can be interpreted is valuable; however, whether or not a statistical outcome suite is accurate, yet constrained by meeting the conditions for stations assumptions of linearity and distribution. While the regularity of complex non-linear interaction structures between multiple sectors comprised of agricultural variables, stalchemy values often involve interactive seasonal variability. On the contrary, ML algorithms, especially ensemble methods (e.g., RF, GBM) and neural networks (e.g., DNN, SVM), have demonstrated considerable capacity for producing models that characterize high-dimensional and non-linear relationships without necessitating extensive feature engineering. For example, while a rule-based irrigation system typically applies water uniformly based on factors such as temperature and time frame, an ML-based irrigation system adaptively adjusts its irrigation schedules based on temperature, real-time soil moisture, weather forecasts, and plant growth stages, which will enable more optimal use of resources. ML also provides higher predictive accuracy on tasks such as yield forecasting, crop health assessment, and disease prediction, and provides more scalable solutions to modern-day agricultural problems. As a result, even though traditional statistical methods or rule-based systems have easier implementation and interpretability characteristics, ML-based smart farming solutions provide more benefits for changing environments, greater reliability and performance, and the ability to scale over time and use in real-world agricultural contexts. Comparative analysis of ML-based smart farming, a rule-based approach, and statistical approaches is included in Table 7.

6.6. Hypothesis testing

Table 8 shows the outcomes of the hypothesis testing relative to the contribution of ML techniques to the domains of agriculture: H1 to H8 represent multiple variables related to the contribution of ML techniques to agriculture: H1 is Resource Optimization and Emission Reduction; H2 is Early Detection and Traditional Practices; H3 is Economic Sustainability; H4 is Interoperability and Job Creation; H5 is Data Privacy and Security; H6 is Climate Resilience and Resource Management; H7 is Soil Health and Erosion Management; and H8 is Water Management and Sustainability. Each of these variables highlights the contributions of ML applications from several points of view. The table shows the associated t-value, β-value, p-value, and significance level for the hypothesis. Each hypothesis tests the contribution of ML techniques in the various aspects of agriculture. The t-value is a measure of how strong the relationship is between the variables being tested. It measures how much the data supports the hypothesis: the higher the absolute t-value, the relationship becomes. The β-value is a coefficient value to assess the strength and direction of the relationship between the independent and dependent variables, while the p-value is a statistical measure to determine if the results are significant.

H1 has a t-value of 8.778, which indicates a strong relationship, a

Table 7
Comparison of ML-Based Smart Farming vs. Rule-Based Systems and Statistical Models.

Criteria	Rule-Based Systems	Statistical Models (e.g., LR)	ML-Based Smart Farming
Adaptability to New Data	Low	Medium	High
Handling Non-linear Data	Low	Low–Medium	High
Predictive Accuracy	~60 %	~70 %	~90 %
Resource Optimization	Basic threshold-based	Moderate (based on variable correlation)	Advanced (real-time, dynamic)
Implementation Cost	Low	Low	Medium–High
Maintenance Needs	Low	Low	Medium
Scalability	Limited	Moderate	High
Interpretability	High	High	Medium
Yield Improvement (est.)	~5–10 %	~10–15 %	~20 %
Water Savings (est.)	~5 %	~10 %	~15 %

Table 8
Hypothesis testing result.

Hypotheses	t-value	β-value	p-value	Result
H1	8.778	0.579	**	Significant
H2	7.629	0.326	**	Significant
H3	6.984	0.340	***	Significant
H4	9.119	0.601	**	Significant
H5	7.893	0.309	*	Significant
H6	6.321	0.295	**	Significant
H7	7.142	0.415	**	Significant
H8	6.456	0.382	*	Significant

Note: ***p < 0.001, **p < 0.01, *p < 0.05.

β-value of 0.579, and a p-value of less than 0.01, indicating a very significant relationship. Hence, ML techniques make a significant contribution to the early detection of growth anomalies and crop diseases. H2 has a t-value of 7.629 with a β-value of 0.326, and the p-value is less than 0.01, indicating a very significant relationship. In this case, ML techniques make a significant contribution to reducing greenhouse gas emissions from agricultural practices. H3 has a p-value of less than 0.001, indicating a highly significant relationship. So, this proves that ML techniques make a significant contribution to reducing soil erosion/land degradation.

For H4, the t-value is 9.119, with a β-value of 0.601, and the p-value <0.01. Thus, the relationship is very significant. For H5, the t-value is 7.893, the β-value of 0.309, and the p-value is < 0.05. This indicates a significant relationship. For H6, the t-value is 6.321, the β-value is 0.295, and the p-value <0.01, meaning it's a very significant relationship. For H7, the t-value is 7.142, the β-value is 0.415, and the p-value <0.01, indicating ML techniques are a very significant contributor to improving economic sustainability in agriculture. For H8, the t-value is 6.456, β-value is 0.382, and p-value <0.05, indicative of a significant relationship as it pertains to ML techniques and promoting water management and sustainability.

Overall, all of the hypotheses measures provided statistically significant relationships between ML techniques and important aspects of agriculture, and the associated significance level reflects the strength of each of these relationships, and reinforces the positive role of ML in agriculture.

7. Discussion

The incorporation of ML into farming is transforming the industry by outlining major challenges such as the optimization of resources, sustainability, and economic resiliency. Traditional practices use historical data and a manual monitoring approach, which creates inefficient resources and source allocations while prescribing uniform applications to input resources that result in peak emissions. Current ML techniques, such as RF and SVM, allow better precision in the analysis of multi-source data that improves input utilization and minimizes environmental impact (Kemmannu et al., 2024; Li et al., 2024). For instance, ML frameworks can optimize cropping patterns and resource allotment at a regional level, which improves long-term sustainable practices (Karimi et al., 2024; Taheri et al., 2024). Formally, traditional systems for disease detection using visual inspections tend to be too late to mitigate the risks of the disease. On the other hand, current ML approaches, primarily CNN, allow for early disease detection via continuously analyzing changes in images, which allows interventions to occur rapidly (Patra et al., 2024; Joshi et al., 2025). Finally, the emergence of DL in a federated system allows for monitoring crops' health without compromising data privacy (Behera et al., 2025). Economically, traditional decision-making is reliant on experience and intuition, which often leads to poor results. Approaches in ML, such as LR and GBM, have powered past traditional approaches rooted in estimating yields and market trends, leading to financially sound decisions based on data (Sahoo et al., 2024; Kheir et al., 2025). Furthermore, explainable ML models and AutoML environments have allayed much of the technology anxiety for non-technical audiences and aided scalable knowledge transfer (Kumar et al., 2025; Paudel et al., 2023). Security is an important challenge in traditional systems that involve vulnerable data sharing. These security risks are mitigated by FL and Differential Privacy, which enable cooperative model training so that farms can develop ML strategies without disclosing their private farm information (Farooq et al., 2022; Friha et al., 2022; Li et al., 2024). Climate resilience is another issue where traditional management makes static predictions using models that will not respond to increasing variability. ML models (e.g., as Gaussian Process Regression, ensemble learning) permit dynamic decisions and resource use, taking climate risk into account and the necessity to respond to climate variability in a resource-limited situation (Lee et al., 2025; Sheline et al., 2024). Soil quality assessments have also moved from generalized methods to precision methods that use Support Vector Regression and hybrid stacking models that improve decision-making regarding irrigation and soil at short spatial scales (Singha et al., 2025; Li et al., 2024). Conventional irrigated production systems often have fixed irrigation schedules that lead to over- or under-irrigation. Methods that support farmers' irrigation decisions based on the application of many ML programs, such as KNN, RNN, and evapotranspiration, will be desirable (Lee et al., 2025; Abuzanouneh et al., 2022; Sheline et al., 2024). The applications of ML in diverse contexts do not only augment the farmer's traditional agricultural practices, they also provide an enhancement of existing practices where farmers apply new timely, accurate, and robust tools to support improved environmental stewardship, productivity, and economic sustainability over the long run - the potency of digital agriculture (Jararweh et al., 2023; Mohanty et al., 2025).

7.1. Theoretical implications

When integrating ML into agriculture, traditional agricultural theories provide a path for reevaluation of existing techniques of resource management, sustainability, and agricultural economics. With the help of a decentralized privacy-preserving framework, the risk obtained by the centralized framework related to data misuse and data breaches is prevented. This shift creates new theoretical perspectives that not only enhance the productivity or efficiency but also prioritize ethical ML use and decision making at the center of data governance. Further, ML invites researchers to think about other new theoretical perspectives that create more dynamic models that consider the relationship between human knowledge, technology, and environmental phenomena. It creates an opportunity to develop interdisciplinary inquiries that consider agriculture in terms of data science, ecology, and economics, to expand agriculture to other areas of inquiry. In addition to moving forward theoretical perspectives in agriculture, the uptake of ML will also prompt questions relating to data ethics, privacy, and the socio-economic consequences of technology adoption for farming communities, considerations that will moving forward inform, and deepen theoretical orientations about how ML will change the future of agriculture though more inclusive, transparent, and decentralized decision-making.

7.2. Practical implications

Numerous practical benefits are obtained using the integration of ML within agriculture that including improved resource optimization and mitigated environmental effects. Informed decisions were made by the farmers with the help of data-driven approaches, which led to cost savings and enhanced productivity. Using ML approaches, the diseases in the plants are identified in the early stage, and it is prevented. This promotes sustainability and reduces the use of chemicals. Also, the inclusion of ML strategies boosts the soil and water management, which leads to long-term agricultural resilience. Another practical aspect of FL and data ownership controls to enhance farmer agency allows farmers to

utilize ML systems without privacy concerns. These features mitigate adoption barriers of trust and transparency, along with ethical concerns, while promoting wider use and greater fidelity toward responsible use of smart digital agricultural technologies, particularly within smallholder and subsistence community contexts. Greater demand for skilled positions in the agriculture industry also offers job opportunities, which promote economic growth in rural agricultural communities. Although Section 6.4 identified significant initial costs in terms of devices and subscriptions, this paper provided some cost-benefit analysis as well as practical mechanisms such as cooperative-style practices, financial subsidies, and public-private partnerships to encourage the adoption of smart farming practices, especially for smallholder farmers. Several practical solutions and support mechanisms are available to help smallholder farmers overcome the affordability and technical challenges of thinking, implementing, and using smart farming technologies. The government offers public financial support mechanisms, such as subsidies and grants, to help offset the initial cost of implementing smart systems. Farmer cooperatives, in many developing locations, are an effective way to share ownership and the expense of smart tools, even in extreme circumstances of poverty. A wide range of micro-financing programs and access to lower-cost credit programs, for example, pay-as-you-go models and low-interest loans, allow farmers to invest in technology over time in affordable amounts. The investment of time and costs into capacity-building programs, conducted by government authorities and NGOs, develops the necessary technical and non-technical skills that will allow farmers to run systems that incorporate ML, regardless of their level of technological skills. Modular or scalable levels of investment in smart farming practices, for example, low-cost soil sensors or mobile applications that make use of ML, allow farmers to access smart agricultural practices in smaller, more manageable parts. Public-private partnerships (PPPs) build relationships between governments, technology companies, and agricultural cooperatives, which further facilitate the adoption of smart farming practices by bundling all aspects of the project together: equipment, training, and ongoing technical support. These solutions are vital in democratizing smart agriculture, and so smallholder farmers, the majority of farmers in many less developed regions, are not locked out of the potential benefits of smart farming and digital agricultural transformation. Smart farming has the potential to promote both economic empowerment and environmental sustainability on a large scale because it lowers barriers to entry.

7.3. Limitation

While this study provided benefits of incorporating ML in agriculture, several challenges still exist. The most significant limitation is that many farmers, particularly in developing regions, do not have the technical skills that enable them to effectively use these technologies. Limited internet connectivity and a lack of appropriate systems to collect and analyze data for agronomic use also limit the uses of these technologies, especially in rural areas. Further, the variable noise of sensor data in real-life adoption, incomplete datasets, and unmatched local environmental and farming attributes lead to a decrease in model accuracy and generalization. Other limitations include issues around ownership of data, fear of exposing sensitive data on crop or yield, which affect adoptability among smallholder farmers who may not have experience with practices related to data privacy. There may also be a fear of job displacement from automation, which may evoke resistance among traditional farmers.

7.4. Future scope

Future research will perform interdisciplinary collaboration between the data science, agriculture, and technology sectors to boost the robustness and applicability of the ML models in agriculture. The collaborative models will also include examining new, scalable solutions that can be addressed in a context of real-world agricultural problems. There will be a focus on training and education to enhance digital literacy for farmers. Educational institutions and agriculture groups can find ways to collaborate to provide targeted workshops and access ML with more applied sessions to give farmers hands-on experience and develop their access to ML capabilities. To make improvements around access, future research partnerships will also provide support in terms of infrastructure, such as access to public wi-fi in rural communities or subsidized internet access packages to foster increased access to data-driven decision-making processes. Additionally, they will look to address issues around data sovereignty, privacy, and income generation for farmers through new farmer-centric data ownership models that promote individual agency and control over data sharing and monetizing. So far, human-in-the-loop approaches have suggested a way to balance the functioning of automated technology with the expert judgement of the human operators, particularly in scenarios of high stakes or ambiguity. ML systems can be prone to failure due to errors in prediction, bias, or generalization. In future work, we will employ methods of outlier detection, missing data imputation, domain adaptation, fairness-aware algorithms, and data augmentation to try to improve robustness. The goal will be to improve resilience in different agricultural contexts and to decrease unintended negative consequences. In terms of generalizability, the datasets used in the existing study are all internal, and in future research, we want to be able to test on independent datasets using public data or more regionally relevant, disparate datasets. This will allow for additional hyperparameter tuning and performance evaluation across both environmental, crop type, and different geographies, ultimately improving the real-world adaptability of ML models. Finally, for decision-making in a real-time way, particularly for time-sensitive decision-making such as irrigation decisions and disease detection, we will work on next-generation ML models that include edge-computing solutions, where smart model codes are sent closer to the location of the data to use local or source data with minimal delay. For effective operation in rural areas, low-power AI models will be evaluated by examining pruning and quantization methods. Low-power AI models, beyond those developed in this research, lightweight neural network architectures, and other efficient communication methods, will enable always-on monitoring in the field via drones, IoT sensors, and smartphones. Finally, future research will expand the economic analysis with a comparative and more granular study with different-sized farms, regions, and crops to gain knowledge of the economic viability, scalability, and adaptability of ML-based farm systems over varying real-world conditions. Future work will involve testing lightweight ML models like decision trees and ensemble methods using local sensor datasets in low-infrastructure rural areas. These findings are especially relevant to emerging economies experiencing parallel challenges such as labor shortages, climate stress, and the digital divide in precision agriculture. Future initiatives will focus on evaluating the performance of energy-efficient ML models in resource-constrained agricultural settings to assess their practicality and scalability. Nevertheless, due to differences in socio-economic, cultural, and infrastructural contexts, comparable studies in other geographic areas are recommended to validate and contextualize the conclusions. Conducting similar research in diverse regions such as Sub-Saharan Africa, Southeast Asia, or Latin America would deepen our global understanding of ML integration in agriculture and inform more localized, equitable policy and technology deployment strategies.

8. Conclusion

The study provides an empirical exploration of ML applications in agriculture using a quantitative research model. The data were collected through a questionnaire administered to 847 farmers. The data were analyzed using SPSS software. The incorporation of ML in agriculture represents a fundamental shift in direction to help overcome significant issues around resource optimization, sustainability, profitability, and environmental concerns. These data-driven decisions help farmers

utilize advanced ML techniques in ways that will direct them to increase their productivity, decrease their costs, and minimize their environmental footprint. This framework illustrates how ML applications, such as early disease detection, increased irrigation management, and optimized use of resources, contribute to a more efficient agricultural environment. Moreover, the promise of job creation and support for economic sustainability reflects the transformative role of ML, securing agricultural practices to be resistant and adaptive. The outcomes demonstrate that variables such as Resource Optimization and Emission Reduction, Early Detection and Traditional Practices, Soil Health and Erosion Management, Economic Sustainability, Interoperability and Job Creation, Data Privacy and Security, Climate Resilience and Resource Management, and Water Management and Sustainability has AVE of 0.671, 0.722, 0.591, 0.626, 0.789, 0.666, 0.653, and 0.700, respectively. It should be noted that the vast majority of farmers currently lack the skills, infrastructure, and trust in using ML for agriculture. Future directions for this area should include training, better connectivity, and ownership of secure data. Governments should subsidize ML training programs for farmers. Stakeholders should invest in rural internet infrastructure. Data privacy laws should be developed to protect farmers' data. Advancing hardware, software, and communication systems will further support adoption. While the research is focused on three agriculturally significant provinces in China, the insights gained, particularly regarding adoption barriers, ethical concerns, and the effectiveness of ML tools, are broadly applicable to other regions with similar agricultural, economic, or climatic conditions.

CRediT authorship contribution statement

Weiye Wang: Writing – original draft, Methodology, Funding acquisition. **Qing Li:** Software, Project administration, Methodology.

Informed consent

Informed consent was obtained from all individual participants included in the study.

Availability of data and material

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Human and animal rights

This article includes studies involving human or animal subjects conducted by the authors.

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Declaration of competing interest

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Data availability

Data will be made available on request.

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